*Remark* 0.1 (Wave-14 reconstitution anchor). This standalone is the inscription of the Platonic holographic-datum heptagon. The closure wave (2026-04-16) added two load-bearing edges to the earlier five-face star. Face (6) is the Drinfeld-centre identification $Z(\mathsf{Rep}_{\mathrm{fact}}(\mathcal{A})) \simeq \mathsf{Rep}_{\mathrm{fact}}(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}))^{\mathsf{E}_2}$ via categorified bar-cobar with half-braiding (cross-ref thm:universal-holography-master, Vol II `chapters/theory/sc_chtop_heptagon.tex`). Face (7) is the derived-algebraic-geometry edge via PTVV 1-shifted symplectic on $\mathrm{Map}(X \times \mathbb{R}_{\geq 0}, B\,\mathrm{SC\text{-}Alg})$. Every face is genuinely independent (no tautological routing); the seven classical identifications (operadic, Koszul, factorization, BV–BRST, convolution, Drinfeld-centre, derived-AG) all factor through the chiral Hochschild cochain complex as hub. The topologisation from $\mathrm{SC}^{\mathrm{ch,top}}$ to $\mathsf{E}_3^{\mathrm{top}}$ is PROVED chain-level on the original complex for classes $G$, $L$, $C$, and the critical level (Vol II `chapters/theory/sc_chtop_heptagon.tex`, thm:topologization-tower); for class $M$ it is weight-completed (correct scope, direct-sum chain-level genuinely fails).

# THE HOLOGRAPHIC MODULAR KOSZUL DATUM

RAEEZ LORGAT

Abstract. Every 3d holomorphic-topological gauge theory $T$ on $\mathbb{R}_+ \times X$ is controlled by a six-component package $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, \mathcal{C}, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$: boundary chiral algebra, Koszul dual, line-operator category, spectral $r$-matrix, universal Maurer–Cartan element, and modular shadow connection. We define each component from the bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ and its single structural identity $D^2_{\mathcal{A}} = 0$, and compute the datum in full for three families: Heisenberg $\mathcal{H}_k$ (abelian Chern–Simons, class $\mathbf{G}$), affine $\widehat{\mathfrak{sl}}_{2k}$ (non-abelian Chern–Simons, class $\mathbf{L}$), and Virasoro $\mathrm{Vir}_c$ (3d gravity, class $\mathbf{M}$). The bulk–boundary–line triangle is recovered from the datum: the bulk is the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, the boundary is $\mathcal{A}$ itself, and the line category is $\mathcal{A}^!$-mod. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure emerges on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ of derived centre and boundary algebra, not on the bar complex. The central organising principle is the $\mathrm{SC}^{\mathrm{ch,top}}$ *heptagon*: seven edges (operadic / Koszul / factorization / BV–BRST / convolution / Drinfeld centre / derived-AG) identify the datum across presentations, with six classical faces and a seventh derived-algebraic-geometry face inscribed via PTVV on $\mathrm{Map}(X \times \mathbb{R}_{\geq 0}, B\,\mathrm{SC\text{-}Alg})$. We construct the anomaly completion: the transgression algebra $B_\Theta$ with secondary anomaly $u = \eta^2 = \kappa(\mathcal{A}^!) \cdot \omega_g$, whose vanishing at the critical central charge is the Clifford/exterior bifurcation governing the genus tower.

## Contents

1. Introduction 2
2. The datum 4
2.1. Boundary algebra 4

2.2. Bar complex 4
2.3. Koszul dual 4
2.4. Line-operator category 5
2.5. Spectral $r$-matrix 5
2.6. Universal Maurer–Cartan element 5
2.7. Shadow connection 6
2.8. Collision filtration and shadow depth 6
3. The Heisenberg datum 6
4. The Kac–Moody datum 7
5. The Virasoro datum 9
6. Comparison table 10
7. The BBL triangle 11
7.1. The three vertices 11
7.2. The three edges 11
7.3. Reconstruction from the datum 12
8. The anomaly completion 12
8.1. The transgression algebra 12
8.2. The secondary anomaly 13
8.3. Genus-$g$ Clifford completion 13
8.4. The dichotomy 13
8.5. The three families 14
8.6. The anomaly parameter as holographic observable 14
9. Concluding remarks 15
9.1. The $E_n$ circle 15
9.2. Scope 15
9.3. From the datum to physics 15
References 15

## 1. Introduction

The problem is to organise the algebraic data of a 3d holomorphic-topological (HT) quantum field theory into a single invariant from which every physical observable reconstructs. A 3d HT theory on the half-space $\mathbb{R}_+ \times X$, where $X$ is a smooth algebraic curve, produces three observation algebras: a boundary chiral algebra $\mathcal{A}$ at $\{0\} \times X$, a bulk algebra in the interior, and a category of topological line operators along $\mathbb{R}_+$. The algebraic engine relating these three is Koszul duality in the homotopical modular chiral realm on algebraic curves: the bar construction on $\mathcal{A}$ produces a coalgebra from which the Koszul dual, the $r$-matrix, and the genus tower are extracted. The governing identity is $D_{\mathcal{A}}^2 = 0$; the governing equation is the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

The holographic modular Koszul datum is

$$\mathcal{H}(T) \;=\; (\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}). \tag{1.1}$$

One six-component package, extracted from a single Maurer–Cartan element by a sequence of projections. The components are:

(i) the boundary chiral algebra $\mathcal{A}$;
(ii) the chiral Koszul dual $\mathcal{A}^! = (H^*(\bar{B}(\mathcal{A})))^\vee$;
(iii) the line-operator category $\mathcal{C} \simeq \mathcal{A}^!$-mod;
(iv) the classical $r$-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$;
(v) the universal Maurer–Cartan element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$;
(vi) the modular shadow connection $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$.

The bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is an $\mathsf{E}_1$-chiral coassociative coalgebra: it carries a differential (from the chiral product) and a deconcatenation coproduct (from the tensor coalgebra structure). The $\mathrm{SC}^{\mathrm{ch,top}}$ structure does not live on $\bar{B}(\mathcal{A})$; it emerges on the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$, where the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ is the SC datum with bulk acting on boundary.

Three families illustrate the full range of the datum.

*Heisenberg* $\mathcal{H}_k$. Abelian Chern–Simons at level $k$. Shadow class $\mathbf{G}$, shadow depth 2. The $r$-matrix is $r(z) = k/z$ (vanishing at $k = 0$); the line category is $\mathsf{Vect}$; the shadow connection is trivial. This is the base case: all six components are scalar.

*Affine* $\widehat{\mathfrak{sl}}_{2k}$. Non-abelian Chern–Simons at level $k$. Shadow class $\mathbf{L}$, shadow depth 3. The $r$-matrix is Yang's $r(z) = k\,\Omega/z$ (trace-form; vanishing at $k = 0$). The line category is $\mathrm{Rep}(U_q(\mathfrak{sl}_2))$ at $q = e^{i\pi/(k+2)}$, recovered from the KZ monodromy. The shadow connection is the KZ connection. Every component is matrix-valued.

*Virasoro* $\mathrm{Vir}_c$. 3d gravity. Shadow class $\mathbf{M}$, infinite depth. The $r$-matrix is $r_c(z) = (c/2)/z^3 + 2T/z$ (cubic and simple poles); the quartic contact invariant $Q^{\mathrm{contact}} = 10/[c(5c+22)] \neq 0$. The Koszul dual is $\mathrm{Vir}_{26-c}$, self-dual at $c = 13$. The genus tower is infinite; gravity does not truncate.

The bulk–boundary–line (BBL) triangle is not an ad hoc diagram connecting three independent objects. It is a single object (the bar complex with its $\mathsf{E}_1$-chiral coassociative structure) viewed from three angles. The three edges are: boundary→bulk (the bar map followed by the trace to the ambient complex), bulk→lines (the complementarity cofibre), and boundary→lines (Koszul duality). Commutativity of the triangle is Theorem A.

The anomaly completion adjoins to the Koszul dual $\mathcal{A}^!$ a degree-1 generator $\eta$ with Clifford (not exterior) relation $\eta^2 = \kappa(\mathcal{A}^!) \cdot \omega_g$. The resulting transgression algebra $B_\Theta$ has a secondary anomaly $u = \eta^2 = \kappa(\mathcal{A}^!) \cdot \omega_g$ that controls the genus tower: when $u \neq 0$ the Clifford algebra is a matrix algebra and genera are Morita-equivalent; when $u = 0$ the handle operators become exterior generators and the theory sees genuine surface topology.

**Conventions.** Grading is cohomological ($|d| = +1$). Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar propagator is $d \log E(z, w)$, weight 1 in both variables regardless of field weight. OPE-mode notation: $a_{(n)}b$. The augmentation ideal is $\bar{\mathcal{A}} = \ker(\varepsilon)$; the bar complex uses $\bar{\mathcal{A}}$, not $\mathcal{A}$.

## 2. The datum

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve $X$.

**Definition 2.1** (The holographic modular Koszul datum)**.** The *holographic modular Koszul datum* of $\mathcal{A}$ is the six-component package

$$\mathcal{H}(\mathcal{A}) \;=\; (\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}), \tag{2.1}$$

with components defined in Definitions 2.2–2.8 below.

### 2.1. Boundary algebra.

**Definition 2.2** (Boundary algebra)**.** The *boundary algebra* is the chiral algebra $\mathcal{A}$ on $X$, restricted from the holomorphic boundary $\{0\}\times X \subset \mathbb{R}_+\times X$ of the 3d HT system.

The boundary carries the full OPE data and is the input to the bar construction. In the 3d HT picture, $\mathcal{A}$ is the boundary condition at the $t=0$ wall: the holomorphic direction produces the OPE, and the topological direction produces the ordering.

### 2.2. Bar complex.

The ordered bar complex is

$$\bar{B}^{\mathrm{ord}}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}), \tag{2.2}$$

where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and $s^{-1}$ denotes desuspension ($|s^{-1}v| = |v|-1$). This is an $\mathsf{E}_1$-chiral coassociative coalgebra: the differential $d_{\bar{B}}$ extracts OPE residues at collision divisors of the Fulton–MacPherson space $\overline{C}_n(\mathbb{C})$, and the deconcatenation coproduct

$$\Delta(a_1\otimes\cdots\otimes a_n) \;=\; \sum_{p=0}^{n}(a_1\otimes\cdots\otimes a_p)\otimes(a_{p+1}\otimes\cdots\otimes a_n) \tag{2.3}$$

splits an ordered tensor sequence at a cut point.

The symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$ is the $\Sigma_n$-coinvariant image under the averaging map $\mathrm{av}\colon \bar{B}^{\mathrm{ord}} \to \bar{B}^{\Sigma}$. The descent from ordered to symmetric is $R$-matrix-twisted: $\bar{B}^{\Sigma}_n = (\bar{B}^{\mathrm{ord}}_n)^{R\text{-}\Sigma_n}$. The ordered bar is the primitive object; the symmetric bar is its coinvariant shadow.

**Warning 2.3** (Bar complex is not an SC-coalgebra)**.** The bar complex $\bar{B}(\mathcal{A})$ is an $\mathsf{E}_1$-chiral coassociative coalgebra. It does *not* carry an $\mathrm{SC}^{\mathrm{ch,top}}$ structure. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure emerges on the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$, computed using the bar complex as a resolution. The pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\mathcal{A})$ is the SC datum.

### 2.3. Koszul dual.

**Definition 2.4** (Chiral Koszul dual)**.** The *chiral Koszul dual* of $\mathcal{A}$ is

$$\mathcal{A}^! \;=\; \big(H^*(\bar{B}(\mathcal{A}))\big)^{\vee}, \tag{2.4}$$

the graded linear dual of the bar cohomology coalgebra $\mathcal{A}^{\mathrm{i}} = H^*(\bar{B}(\mathcal{A}))$.

The passage from $\mathcal{A}$ to $\mathcal{A}^!$ involves three steps: bar construction ($\mathcal{A} \mapsto \bar{B}(\mathcal{A})$, coalgebra), cohomology ($\bar{B}(\mathcal{A}) \mapsto \mathcal{A}^{\mathrm{i}}$, dual coalgebra), and Verdier duality ($\mathcal{A}^{\mathrm{i}} \mapsto \mathcal{A}^! = (\mathcal{A}^{\mathrm{i}})^\vee$, dual algebra). These are three distinct functors producing four distinct objects:

$$\mathcal{A} \quad \longmapsto \quad \bar{B}(\mathcal{A}) \quad \longmapsto \quad \mathcal{A}^{\mathrm{i}} = H^*(\bar{B}(\mathcal{A})) \quad \longmapsto \quad \mathcal{A}^! = (\mathcal{A}^{\mathrm{i}})^\vee. \tag{2.5}$$

The cobar construction $\Omega(\bar{B}(\mathcal{A})) \simeq \mathcal{A}$ is *inversion*, not Koszul duality.

### 2.4. **Line-operator category.**

**Definition 2.5** (Line-operator category)**.** The *line-operator category* is

$$\mathcal{C} \;\simeq\; \mathcal{A}^!\text{-}\mathsf{mod}. \tag{2.6}$$

Objects are topological defect lines along the $\mathbb{R}_+$-direction; morphisms are junction operators. The braiding on $\mathcal{C}$ comes from crossing in the holomorphic plane: two line operators separated in $\mathbb{C}_z$ can be exchanged by moving one around the other. On the Koszul locus, the $r$-matrix $r(z)$ determines the braiding explicitly.

### 2.5. **Spectral $r$-matrix.**

**Definition 2.6** (Classical $r$-matrix)**.** The *spectral $r$-matrix* is the binary genus-0 collision residue

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \;\in\; \mathcal{A}^! \,\widehat{\otimes}\, \mathcal{A}^![\![z^{-1}, z]\!]. \tag{2.7}$$

The $r$-matrix satisfies the classical Yang–Baxter equation (CYBE), which follows from the Arnold relation on $\overline{M}_{0,4}$: the three boundary divisors of the genus-0 4-pointed moduli space correspond to channels (12|34), (13|24), (14|23), and evaluating the Arnold relation on $\Omega \otimes \Omega \in (\mathcal{A}^!)^{\otimes 3}$ produces the three terms of $\sum [r_{ij}, r_{ik}] = 0$.

The $r$-matrix lives in the $\mathsf{E}_1$ convolution algebra $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$; the modular characteristic $\kappa(\mathcal{A})$ is its $\Sigma_2$-coinvariant projection. For abelian algebras, $\mathrm{av}(r(z)) = \kappa$ directly. For non-abelian affine Kac–Moody algebras, the full modular characteristic includes the Sugawara shift:

$$\kappa(V_k(\mathfrak{g})) \;=\; \mathrm{av}(r(z)) + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}, \tag{2.8}$$

where the $\dim(\mathfrak{g})/2$ term is the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$.

### 2.6. **Universal Maurer–Cartan element.**

**Definition 2.7** (Universal MC element)**.** The *universal Maurer–Cartan element* is

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d_0 \;\in\; \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}), \tag{2.9}$$

where $D_{\mathcal{A}}$ is the full bar differential and $d_0$ is its linear (genus-0, degree-1) part.

The element $\Theta_{\mathcal{A}}$ is always MC because $D_{\mathcal{A}}^2 = 0$:

$$D\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0. \tag{2.10}$$

This is the bar-intrinsic construction: $\Theta_{\mathcal{A}}$ exists for every chirally Koszul algebra, at all degrees, as an inverse-limit $\Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$. The modular characteristic, the $r$-matrix, the shadow connection, and the full genus tower are projections of $\Theta_{\mathcal{A}}$.

2.7. **Shadow connection.**

**Definition 2.8** (Shadow connection)**.** The *modular shadow connection* on $\overline{\mathcal{M}}_{g,n}$ is

$$\nabla_{g,n}^{\mathrm{hol}} \;=\; d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}}), \tag{2.11}$$

where $\mathrm{Sh}_{g,n}$ is the shadow projection to the $(g,n)$-component of the modular convolution algebra.

Flatness of $\nabla^{\mathrm{hol}}$ is the MC equation projected to $(g,n)$. At genus 0, $\nabla_{0,n}^{\mathrm{hol}}$ is unconditionally flat; at genus $g \geq 1$, the connection acquires curvature $\kappa(\mathcal{A}) \cdot \omega_g$ (the genus-$g$ anomaly).

The shadow connection encodes the full genus tower. Its monodromy representations are the physical amplitudes: at genus 0, the KZ monodromy for affine algebras; the BPZ monodromy for Virasoro.

2.8. **Collision filtration and shadow depth.** The modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ carries a collision filtration $F_{\mathrm{coll}}^0 \supset F_{\mathrm{coll}}^1 \supset \cdots$ indexed by minimum pairwise distance on $\overline{C}_n(\mathbb{C})$. Depth 0 is free propagation; depth $k$ is $k$-fold nested collision. The spectral sequence $E_1^{p,*} = H^*(\mathrm{gr}_{\mathrm{coll}}^p\, \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}, d_0)$ converges to $H^*(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$. Collapse is governed by shadow depth:

| Class | Shadow depth | OPE pole order | Representative |
|---|---|---|---|
| **G** | 2 | 2 | Heisenberg $\mathcal{H}_k$ |
| **L** | 3 | 3 | Affine $\widehat{\mathfrak{g}}_k$ |
| **C** | 4 | 4 | $\beta\gamma$ system |
| **M** | $\infty$ | $\geq 4$ | Virasoro, $\mathcal{W}_N$ |

The discriminant $\Delta = 8\kappa S_4$ controls truncation: $\Delta = 0$ (classes **G**, **L**, **C**) gives a finite shadow tower; $\Delta \neq 0$ (class **M**) gives an infinite tower. The discriminant is linear in $\kappa$, not quadratic.

## 3. The Heisenberg datum

The Heisenberg algebra $\mathcal{H}_k$ at level $k$ is the abelian current algebra generated by a single field $J$ with OPE

$$J(z)\, J(w) \;\sim\; \frac{k}{(z-w)^2}. \tag{3.1}$$

This is abelian Chern–Simons for $G = U(1)$ at level $k$.

**Proposition 3.1** (Heisenberg datum)**.** *The holographic datum of $\mathcal{H}_k$ is*

$$\mathcal{H}(\mathcal{H}_k) \;=\; \big(\mathcal{H}_k,\ \mathrm{Sym}^{\mathrm{ch}}(V^*),\ \mathsf{Vect},\ k/z,\ k\cdot\eta\otimes\Lambda,\ d\big). \tag{3.2}$$

*Proof.* We verify each component.

*(i) Boundary.* $\mathcal{A} = \mathcal{H}_k$, the free boson at level $k$.

*(ii) Koszul dual.* $\mathcal{A}^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$. This is the curved symmetric chiral algebra with $m_0 = -k\omega$; it is *not* $\mathcal{H}_{-k}$ (the same algebra has the same $\kappa$ on both sides, but the duals differ as curved objects).

*(iii) Lines.* $\mathcal{C} = \mathsf{Vect}$. The line category is trivial: abelian gauge theory has no nontrivial Wilson lines.

*(iv) $r$-matrix.* $r(z) = k/z$. The OPE has a double pole; $d\log$ absorbs one, leaving a simple pole. At $k = 0$: $r = 0$ (abelian limit).

*(v) MC element.* $\Theta_{\mathcal{H}_k} = k\cdot\eta\otimes\Lambda$, where $\eta = d\log(z_1 - z_2)$ is the Arnold form and $\Lambda$ is the unique (up to scale) bilinear on $V$. The MC equation is satisfied because $\mathcal{H}_k$ is class $\mathbf{G}$: all operations beyond degree 2 vanish.

*(vi) Shadow connection.* $\nabla^{\mathrm{hol}} = d$. The shadow connection is flat and trivial: the MC element is scalar, so $\mathrm{Sh}_{g,n}(\Theta_{\mathcal{H}_k})$ is a multiple of the identity for all $(g, n)$. □

*Remark* 3.2 (Modular characteristic)*.* The modular characteristic is $\kappa(\mathcal{H}_k) = k$. This is the abelian prototype: $\mathrm{av}(r(z)) = \mathrm{av}(k/z) = k = \kappa$. The Koszul complementarity is $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = 0$ (class $\mathbf{G}/\mathbf{L}$/free-field complementarity).

*Remark* 3.3 (Quantised $R$-matrix)*.* The quantised $R$-matrix is $R(z) = \exp(k\hbar/z)$, scalar and automatically satisfying the Yang–Baxter equation. The monodromy $\exp(-2\pi i k)$ is the Aharonov–Bohm phase.

## 4. The Kac–Moody datum

The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level $k$ is generated by currents $J^a(z)$ ($a = 1, \ldots, \dim(\mathfrak{g})$) with OPE

$$J^a(z)\,J^b(w) \;\sim\; \frac{k\,\kappa^{ab}}{(z-w)^2} \;+\; \frac{f^{ab}{}_c\,J^c(w)}{z-w}, \tag{4.1}$$

where $\kappa^{ab}$ is the Killing form and $f^{ab}{}_c$ are structure constants. This is 3d Chern–Simons for $G$ at level $k$. We take $\mathfrak{g} = \mathfrak{sl}_2$ for concreteness; the formulae generalise to arbitrary simple $\mathfrak{g}$.

**Proposition 4.1** (Affine $\widehat{\mathfrak{sl}}_2$ datum)**.** *The holographic datum of $\widehat{\mathfrak{sl}}_{2k}$ is*

$$\mathcal{H}(\widehat{\mathfrak{sl}}_{2k}) \;=\; \Big(\widehat{\mathfrak{sl}}_{2k},\ \widehat{\mathfrak{sl}}_{2-k-4},\ \mathcal{O}_q^{\mathrm{sh}},\ \frac{k\,\Omega}{z},\ \Theta_k,\ d - \kappa_k\,\omega_g - \cdots\Big), \tag{4.2}$$

*with $q = e^{i\pi/(k+2)}$ and $\kappa_k = \dim(\mathfrak{sl}_2)(k + h^\vee)/(2h^\vee) = 3(k+2)/4$.*

*Proof.* We verify each component.

*(i) Boundary.* $\mathcal{A} = \widehat{\mathfrak{sl}}_{2k}$, the affine Kac–Moody vertex algebra at level $k$. The three generators $J^a$ ($a \in \{e, f, h\}$) produce a triple-pole OPE (double pole from the Killing form, simple pole from the Lie bracket).

*(ii) Koszul dual.* $\mathcal{A}^! = \widehat{\mathfrak{sl}}_{2-k-4}$. The Koszul involution acts by $k \mapsto -k - 2h^\vee$ with $h^\vee(\mathfrak{sl}_2) = 2$. The complementarity sum is $\kappa(\widehat{\mathfrak{sl}}_{2k}) + \kappa(\widehat{\mathfrak{sl}}_{2-k-4}) = 0$, verified at $k = 0$: $\kappa_0 = 3/2$ and $\kappa_{-4} = -3/2$.

*(iii) Lines.* $\mathcal{C} = \mathcal{O}_q^{\mathrm{sh}} \simeq \mathrm{Rep}(U_q(\mathfrak{sl}_2))$. The braided tensor category of line operators is the representation category of the quantum group at $q = e^{i\pi/(k+2)}$. The identification is the affine Drinfeld–Kohno theorem: the KZ monodromy representation on conformal blocks agrees with the quantum-group $R$-matrix action on tensor products of evaluation modules.

*(iv) $r$-matrix.* $r(z) = k\,\Omega_{\mathrm{tr}}/z$ in trace-form convention, where $\Omega_{\mathrm{tr}}$ is the trace-form Casimir. At $k = 0$: $r = 0$ (abelian limit). In Kazhdan–Lusztig (KZ) normalisation: $r(z) = \Omega_{\mathrm{KZ}}/((k + h^\vee)z)$ with $\Omega_{\mathrm{KZ}} = 2h^\vee\,\Omega_{\mathrm{tr}}$ the Killing-normalised Casimir. The two conventions are related by a *level-dependent rescaling of the Casimir*, not by the naive identity $k\,\Omega_{\mathrm{tr}} = \Omega_{\mathrm{KZ}}/(k + h^\vee)$: the latter fails as a rational-function equation in $k$ (at $k = 0$ the left side vanishes, the right side is finite and non-zero). The correct statement is that $r_{\mathrm{tr}}$ and $r_{\mathrm{KZ}}$ are two presentations of the same underlying operadic datum in different bases, exchanged by $\Omega \mapsto \Omega$ and $k \mapsto k/((k + h^\vee) \cdot 2h^\vee)$. At $k = -h^\vee = -2$: trace-form finite, KZ diverges (Sugawara singularity at critical level).

*(v) MC element.* $\Theta_k = \Theta_{\widehat{\mathfrak{sl}}_{2k}}$ is determined by $D_{\mathcal{A}}^2 = 0$. The shadow tower terminates at depth 3 (class $\mathbf{L}$): the quartic invariant $S_4 = 0$ (the discriminant $\Delta = 8\kappa S_4 = 0$), so the shadow obstruction tower is finite.

*(vi) Shadow connection.* At genus 0, degree $n$, the shadow connection is the KZ connection:

$$\nabla_{0,n}^{\mathrm{hol}} \;=\; d - \sum_{i<j} r_{ij}\, dz_{ij} \;=\; d - \frac{1}{k + h^\vee} \sum_{i<j} \frac{\Omega_{ij}\, dz_{ij}}{z_{ij}}, \tag{4.3}$$

where $dz_{ij} = d(z_i - z_j)$. Flatness of the KZ connection is the MC equation at genus 0, degree $n$. At genus $g \geq 1$, the connection acquires curvature $\kappa_k \cdot \omega_g$ with $\kappa_k = 3(k + 2)/4$. □

*Remark* 4.2 (Quantum group line category). The identification $\mathcal{C} \simeq \mathrm{Rep}(U_q(\mathfrak{sl}_2))$ is a reconstruction theorem: the monodromy of the genus-0 shadow connection $\nabla_{0,n}^{\mathrm{hol}}$ (the KZ connection) generates a braided tensor category that is equivalent to the representation category of the quantum group. The spectral parameter $z$ in $r(z) = k\,\Omega/z$ is the residual parameter after $d\log$-absorption from the OPE; on evaluation modules, it becomes the additive spectral parameter of the Yangian. The full quantised $R$-matrix $R(z) = 1 + \hbar\, r(z) + O(\hbar^2)$ satisfies the quantum Yang–Baxter equation.

*Remark* 4.3 (Critical level). At $k = -h^\vee = -2$: $\kappa = 0$, all monodromy is trivial, $H^1$ doubles (from 4 to 8 for $\mathfrak{sl}_2$), and Koszulness fails. The Feigin–Frenkel centre is the algebra of $\mathfrak{sl}_2$-opers on the formal disk: $Z(V_{-2}(\mathfrak{sl}_2)) \simeq \mathrm{Fun}(\mathrm{Op}_{\mathfrak{sl}_2}(D))$. The shadow connection degenerates, and the line category jumps from semisimple to abelian.

## 5. The Virasoro datum

The Virasoro algebra $\mathrm{Vir}_c$ at central charge $c$ is generated by the stress tensor $T$ with OPE

$$T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} \;+\; \frac{2T(w)}{(z-w)^2} \;+\; \frac{\partial T(w)}{z-w}. \tag{5.1}$$

This is 3d gravity: $SL(2,\mathbb{R}) \times SL(2,\mathbb{R})$ Chern–Simons at levels $k_\pm = \ell/(4G)$, with $c = 3\ell/(2G)$ (Brown–Henneaux).

**Proposition 5.1** (Virasoro datum)**.** *The holographic datum of* $\mathrm{Vir}_c$ *is*

$$\mathcal{H}(\mathrm{Vir}_c) \;=\; \big(\mathrm{Vir}_c,\ \mathrm{Vir}_{26-c},\ \mathcal{C}_c,\ r_c(z),\ \Theta_c,\ \nabla_c^{\mathrm{hol}}\big), \tag{5.2}$$

*with*

$$r_c(z) \;=\; \frac{c/2}{z^3} + \frac{2T}{z}, \tag{5.3}$$

$$\kappa_c \;=\; \frac{c}{2}, \tag{5.4}$$

$$\nabla_{1,1}^{\mathrm{hol}} \;=\; d \;-\; \kappa_c\,\omega_1 \;-\; \frac{10}{c(5c+22)}\,\delta_4 \;-\; \cdots\,. \tag{5.5}$$

*Proof.* We verify each component.

*(i) Boundary.* $\mathcal{A} = \mathrm{Vir}_c$, the Virasoro vertex algebra.

*(ii) Koszul dual.* $\mathcal{A}^! = \mathrm{Vir}_{26-c}$. The Koszul involution is $c \mapsto 26 - c$. The complementarity sum is $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = c/2 + (26-c)/2 = 13$. Self-dual at $c = 13$ (not $c = 26$).

*(iii) Lines.* $\mathcal{C}_c$ is the line-operator category, determined by the $r$-matrix monodromy. For the Virasoro algebra, the line category is infinite-depth: the monodromy representation is determined by the full BPZ differential equations, not by a finite-dimensional quantum group.

*(iv) $r$-matrix.* The $r$-matrix $r_c(z) = (c/2)/z^3 + 2T/z$ has a cubic pole (gravitational, from the quartic OPE pole after $d\log$-absorption) and a simple pole (stress-tensor exchange). The quartic contact invariant

$$Q^{\mathrm{contact}} \;=\; \frac{10}{c(5c+22)} \;\neq\; 0 \tag{5.6}$$

is nonzero for all $c \neq 0$, $c \neq -22/5$: this is the obstruction to extending $r(z)$ from degree 2 to degree 4, and its non-vanishing is the signature of class **M**.

*(v) MC element.* $\Theta_c = \Theta_{\mathrm{Vir}_c}$ is the infinite-depth MC element. The shadow tower does not truncate: all shadow invariants $S_r$ ($r \geq 2$) are nonzero. The modular characteristic is $\kappa(\mathrm{Vir}_c) = S_2 = c/2$.

*(vi) Shadow connection.* The shadow connection (5.5) has leading curvature $\kappa_c\,\omega_g = (c/2)\,\omega_g$ at genus $g$, plus an infinite series of corrections from the shadow obstruction tower. □

*Remark* 5.2 (The gravitational hierarchy). Shadow class **M** means the $A_\infty$ tower $(m_1, m_2, m_3, \ldots)$ is infinite: every higher operation $m_k$ is nonzero. The generating depth is $d_{\mathrm{gen}} = 3$ (the operation $m_3$ determines all $m_k$ recursively via the MC equation), but the algebraic depth is $d_{\mathrm{alg}} = \infty$ (no operation vanishes). Gravity does not truncate.

*Remark* 5.3 (Two critical values of $c$). The two critical central charges play distinct roles:

(i) $c = 13$: Koszul self-duality ($\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$). The complementarity parameter is $\kappa + \kappa' = 13 \neq 0$; self-duality is a symmetry enhancement within the perturbative regime.

(ii) $c = 26$: anomaly cancellation ($\kappa(\mathrm{Vir}_{26-c}^!) = (26-c)/2 = 0$ at $c = 26$). The Clifford/exterior bifurcation (Section 8). This is the bosonic string.

Self-duality and anomaly cancellation are separated by 13 units of central charge.

*Remark* 5.4 (Primitive coproduct). The transferred coproduct $\Delta_z^{\mathrm{Vir}}$ is strictly primitive at all degrees: the ghost-number obstruction intrinsic to Drinfeld–Sokolov reduction kills every higher-product-lattice tree correction. All gravitational dynamics resides in the collision residue and the infinite $A_\infty$ product tower; the coproduct channel is identically zero.

## 6. Comparison table

| Component | $\mathcal{H}_k$ | $\widehat{\mathfrak{sl}}_{2k}$ | $\mathrm{Vir}_c$ |
|---|---|---|---|
| $\mathcal{A}$ (boundary) | $\mathcal{H}_k$ | $\widehat{\mathfrak{sl}}_{2k}$ | $\mathrm{Vir}_c$ |
| $\mathcal{A}^!$ (Koszul dual) | $\mathrm{Sym}^{\mathrm{ch}}(V^*)$ | $\widehat{\mathfrak{sl}}_{2-k-4}$ | $\mathrm{Vir}_{26-c}$ |
| $\mathcal{C}$ (lines) | $\mathsf{Vect}$ | $\mathrm{Rep}(U_q(\mathfrak{sl}_2))$ | $\mathcal{C}_c$ (BPZ) |
| $r(z)$ | $k/z$ | $k\,\Omega/z$ | $(c/2)/z^3 + 2T/z$ |
| $\kappa$ | $k$ | $3(k+2)/4$ | $c/2$ |
| $\kappa + \kappa'$ | 0 | 0 | 13 |
| Class | **G** | **L** | **M** |
| Depth | 2 | 3 | $\infty$ |
| $Q^{\mathrm{contact}}$ | 0 | 0 | $10/[c(5c+22)]$ |
| $\nabla^{\mathrm{hol}}$ | $d$ (trivial) | KZ | BPZ (infinite) |
| Physics | abelian CS | non-abelian CS | 3d gravity |

The three families represent three qualitatively distinct regimes. Class **G** is Gaussian: scalar, trivial lines, the datum collapses to a single number $k$. Class **L** is Lie-theoretic: matrix-valued, finite quantum-group lines, the datum is determined by $\mathfrak{g}$ and $k$. Class **M** is gravitational: infinite depth, no finite quantum group, the datum requires the full infinite tower $\Theta_c$.

## 7. The BBL triangle

The bulk–boundary–line (BBL) triangle is the central diagram of 3d holomorphic-topological gauge theory. It is not built from three independent objects connected by ad hoc functors; it is a single object (the bar complex with its $\mathsf{E}_1$-chiral coassociative structure) viewed from three angles.

### 7.1. **The three vertices.**

**Definition 7.1** (BBL triangle)**.** Let $\mathcal{A}$ be a chirally Koszul vertex algebra. The *bulk–boundary–line triangle* of $\mathcal{A}$ consists of:

(i) *Boundary*: the chiral algebra $\mathcal{A}$ itself, the input to the bar construction.

(ii) *Bulk*: the chiral derived centre

$$\mathcal{A}_{\mathrm{bulk}} \;\simeq\; \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \;=\; C^{\bullet}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}), \tag{7.1}$$

computed by chiral Hochschild cochains (not topological Hochschild; the geometry of the curve determines which chiral/topological/categorical Hochschild theory applies).

(iii) *Lines*: the category $\mathcal{C} \simeq \mathcal{A}^!\text{-}\mathsf{mod}$ of modules over the chiral Koszul dual.

The $\mathrm{SC}^{\mathrm{ch,top}}$ structure lives on the *pair* $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\mathcal{A})$: the chiral derived centre carries brace operations and a chiral Gerstenhaber bracket, and the boundary algebra $\mathcal{A}$ is acted upon by the bulk via the universal brace. The SC datum is two-coloured with directionality: bulk-to-boundary only, no boundary-to-bulk.

### 7.2. **The three edges.**

(i) *Boundary* $\to$ *Bulk* (the bar map). The bar construction maps $\mathcal{A}$ to $\bar{B}(\mathcal{A})$; the trace map $\mathrm{tr}_g\colon \bar{B}^{(g)}(\mathcal{A}) \to \mathbf{C}_g(\mathcal{A})$ projects the bar coalgebra into the ambient complex $\mathbf{C}_g(\mathcal{A}) = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}(\mathcal{A}))$. The image $\mathbf{Q}_g(\mathcal{A}) = \mathrm{im}(\mathrm{tr}_g)$ is one half of the complementarity decomposition.

(ii) *Bulk* $\to$ *Lines* (the complementarity cofibre). The complementarity theorem gives

$$\mathbf{C}_g(\mathcal{A}) \;\simeq\; \mathbf{Q}_g(\mathcal{A}) \;\oplus\; \mathbf{Q}_g(\mathcal{A}^!).$$

The cofibre $\mathrm{cofib}(\mathbf{Q}_g(\mathcal{A}) \hookrightarrow \mathbf{C}_g(\mathcal{A})) \simeq \mathbf{Q}_g(\mathcal{A}^!)$ is the dual shadow, controlling the line-operator category through the centre–module adjunction.

(iii) *Boundary* $\to$ *Lines* (Koszul duality). The composite $\mathcal{A} \mapsto \bar{B}(\mathcal{A}) \mapsto H^*(\bar{B}(\mathcal{A})) \mapsto (H^*(\bar{B}(\mathcal{A})))^\vee = \mathcal{A}^!$ is bar-dualize. The module category $\mathcal{A}^!$-$\mathsf{mod}$ is the categorical output.

Commutativity of the triangle is Theorem A (the Verdier intertwining at the algebra level).

7.3. **Reconstruction from the datum.** The datum $\mathcal{H}(\mathcal{A})$ is a complete invariant.

**Proposition 7.2** (Recovery theorems)**.** *The following reconstructions hold:*

(R1) Collision residue is twisting. $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) = \tau|_{\bar{B}^2(\mathcal{A})}$ *(the universal twisting morphism restricted to bar degree* 2*).*

(R2) CYBE from Arnold. *The three boundary divisors of* $\overline{M}_{0,4}$ *produce the three terms of* $\sum [r_{ij}, r_{ik}] = 0$.

(R3) Affine shadow = KZ. *For* $\mathcal{A} = \widehat{\mathfrak{g}}_k$*:* $\mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k}) = (k+h^\vee)^{-1} \sum_{i<j} \Omega_{ij}\, dz_{ij}/z_{ij}$.

(R4) Quartic as $L_\infty$ obstruction. $[\mathfrak{R}^{\mathrm{mod}}_4(\mathcal{A})] = [o_4(\mathcal{A})] \in H^2(\mathcal{A}^{\mathrm{sh}}_{4,0})$ *is the obstruction to extending* $r(z)$ *from degree* 2 *to degree* 4*. Vanishes for* $\mathbf{G}$*,* $\mathbf{L}$*,* $\mathbf{C}$*; nonzero for* $\mathbf{M}$*.*

*Remark* 7.3 (BBL for Heisenberg)*.* Boundary = $\mathcal{H}_k$. Bulk = $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k)$, which is 3-dimensional (the naive centre is 1-dimensional; the derived centre sees $\mathrm{Ext}^1$ and $\mathrm{Ext}^2$). Lines = $\mathsf{Vect}$. The triangle degenerates: the bulk-to-lines edge is trivial, and the boundary-to-lines edge is trivial.

*Remark* 7.4 (BBL for Kac–Moody)*.* Boundary = $\widehat{\mathfrak{sl}}_{2k}$. Bulk = $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\widehat{\mathfrak{sl}}_{2k})$, with $\mathrm{ChirHoch}^1(V_k(\mathfrak{sl}_2)) \cong \mathfrak{sl}_2$ and total dimension $\dim(\mathfrak{sl}_2) + 2 = 5$. Lines = $\mathrm{Rep}(U_q(\mathfrak{sl}_2))$. The Drinfeld–Kohno theorem makes the boundary-to-lines edge constructive: the monodromy of the KZ connection generates $\mathrm{Rep}(U_q(\mathfrak{sl}_2))$. The bulk-to-lines edge is the complementarity cofibre: $\mathbf{Q}_g(\widehat{\mathfrak{sl}}_{2-k-4})$ controls the line-operator anomaly.

*Remark* 7.5 (BBL for Virasoro)*.* Boundary = $\mathrm{Vir}_c$. Bulk = $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$: chiral Hochschild cohomology concentrated in degrees $\{0, 1, 2\}$ with polynomial Hilbert series (this is amplitude, not virtual dimension). Lines = $\mathcal{C}_c$: the infinite-depth line category, governed by the BPZ equations. The triangle is fully active at all three vertices; the bulk-to-lines edge is controlled by the Virasoro complementarity $\mathbf{Q}_g(\mathrm{Vir}_c) \oplus \mathbf{Q}_g(\mathrm{Vir}_{26-c})$.

## 8. The anomaly completion

8.1. **The transgression algebra.** Fix a chirally Koszul algebra $\mathcal{A}$ with Koszul dual $B = \mathcal{A}^!$. The curvature of the bar complex on a genus-$g$ curve is $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$. The Koszul dual carries dual curvature $\lambda = \kappa(B) \cdot \omega_g$.

**Definition 8.1** (Transgression algebra)**.** The *transgression algebra* is

$$B_\lambda \;=\; B\langle\eta\rangle \,/\, (\eta^2 - \lambda), \qquad |\eta| = 1, \quad d_{B_\lambda}(\eta) = 0, \tag{8.1}$$

where $\eta$ is a degree-1 generator with Clifford (not exterior) relation $\eta^2 = \lambda = \kappa(B) \cdot \omega_g$, and the differential on $B$ is extended by $d_{B_\lambda}(b) = d_B(b) + [\eta, b]$ for $b \in B$.

The relation $\eta^2 = \lambda$ is Clifford when $\lambda \neq 0$ and exterior when $\lambda = 0$. This is the algebraic encoding of the anomaly: $\lambda = 0$ means the theory is anomaly-free, and the algebra structure changes qualitatively.

### 8.2. **The secondary anomaly.**

**Definition 8.2** (Secondary anomaly)**.** The *secondary anomaly* is

$$u \;=\; \eta^2 \;=\; \lambda \;=\; \kappa(\mathcal{A}^!) \cdot \omega_g. \tag{8.2}$$

It is closed, central, and acts as the scalar $\kappa(\mathcal{A}^!)$ on standard modules.

The secondary anomaly is the bifurcation parameter for the genus tower. Its behaviour depends on the Koszul complementarity:

| Family | $\kappa(\mathcal{A}^!)$ | $u$ | Regime |
|---|---|---|---|
| $\mathcal{H}_k$ | $-k$ | $-k\,\omega_g$ | $u = 0$ at $k = 0$ |
| $\widehat{\mathfrak{sl}}_{2k}$ | $3(-k-2)/4$ | $3(-k-2)\omega_g/4$ | $u = 0$ at $k = -2$ (critical) |
| $\mathrm{Vir}_c$ | $(26-c)/2$ | $(26-c)\omega_g/2$ | $u = 0$ at $c = 26$ (string) |

### 8.3. **Genus-$g$ Clifford completion.**

**Construction 8.3** (Clifford completion)**.** At genus $g$, adjoin $2g$ handle operators $\alpha_1, \beta_1, \ldots, \alpha_g, \beta_g$ with

$$\alpha_i\beta_j + \beta_j\alpha_i = \delta_{ij}\,u, \qquad \alpha_i\alpha_j + \alpha_j\alpha_i = 0, \qquad \beta_i\beta_j + \beta_j\beta_i = 0. \tag{8.3}$$

The genus-$g$ Clifford completion is the algebra

$$\mathfrak{G}_g(\mathcal{A}) \;=\; B_\lambda \otimes \mathrm{Cl}(H_1(\Sigma_g); u). \tag{8.4}$$

The handle operators record the genus-$g$ topology. Their anticommutator is controlled by the secondary anomaly $u$, which splits quantum field theory into two qualitative regimes.

### 8.4. **The dichotomy.**

**Proposition 8.4** (Clifford/exterior dichotomy)**.** *The genus tower of $\mathcal{A}$ bifurcates at $u = 0$:*

*(i) When $u \neq 0$: the Clifford algebra $\mathrm{Cl}(H_1(\Sigma_g); u)$ is a matrix algebra (dimension $2^g$) after inverting $u$. Genus-$g$ is Morita equivalent to genus-0. All genus corrections are resummable from genus-0 data. This is the perturbative regime.*

*(ii) When $u = 0$: the Clifford anticommutator vanishes. The handle operators become exterior generators on $H_1(\Sigma_g)$ (dimension $2^{2g}$). The theory sees genuine surface topology. The genus tower collapses ($\kappa_{\mathrm{eff}} = 0$).*

*Proof.* Standard Clifford algebra theory. When the bilinear form $u \cdot \langle \cdot, \cdot \rangle$ on $H_1(\Sigma_g, \mathbb{Z})$ is nondegenerate (i.e., $u \neq 0$), the Clifford algebra is a matrix algebra $M_{2^g}(\mathbb{C})$ (even-dimensional, nondegenerate symplectic). When $u = 0$, the bilinear form degenerates and the Clifford algebra becomes the exterior algebra $\Lambda^*(H_1(\Sigma_g)^*)$. □

8.5. **The three families.** *Heisenberg.* $u = -k\,\omega_g$. At $k = 0$: $u = 0$, exterior regime. At generic $k$: $u \neq 0$, perturbative. The bifurcation is at the abelian limit.

*Kac–Moody.* $u = 3(-k-2)\omega_g/4$. At $k = -2$ (critical level): $u = 0$, the Feigin–Frenkel centre appears, the genus tower collapses. At generic $k$: $u \neq 0$, perturbative. The critical level is the bifurcation point.

*Virasoro.* $u = (26-c)\omega_g/2$. At $c = 26$: $u = 0$, the bosonic string becomes topological ($Z_{\mathrm{grav}} = 1$). At generic $c$: $u \neq 0$, perturbative. At $c = 13$: $u = 13\omega_g/2 \neq 0$, so Koszul self-duality lives within the perturbative regime.

The gravitational partition function at genus $g$ is the super-trace of the MC element through the Clifford factor:

$$Z_g \;=\; \mathrm{str}_{\mathfrak{G}_g/B_\Theta}(\Theta_c^{(g)}). \tag{8.5}$$

The complementarity theorem gives a Lagrangian decomposition

$$H^*(\overline{\mathcal{M}}_g,\, \mathcal{Z}(\mathrm{Vir}_c)) \;\simeq\; \mathbf{Q}_g(\mathrm{Vir}_c) \;\oplus\; \mathbf{Q}_g(\mathrm{Vir}_{26-c}) \tag{8.6}$$

into boundary (+1-eigenspace of Verdier involution) and bulk (−1-eigenspace) sectors. The shifted symplectic structure makes both Lagrangian.

8.6. **The anomaly parameter as holographic observable.** The secondary anomaly $u$ is the single scalar that governs the passage from genus 0 to all genera. For each family, $u$ vanishes at a distinguished value of the level/central charge:

| Family | $u = 0$ at | Physical meaning | Regime |
|---|---|---|---|
| $\mathcal{H}_k$ | $k = 0$ | trivial theory | exterior |
| $\widehat{\mathfrak{g}}_k$ | $k = -h^\vee$ | critical level, Feigin–Frenkel | exterior |
| $\mathrm{Vir}_c$ | $c = 26$ | bosonic string, anomaly cancellation | exterior |

The datum $\mathcal{H}(\mathcal{A})$ together with the anomaly completion $B_\Theta$ controls the full 3d HT correspondence: the six components determine the genus-0 physics (the datum), and the secondary anomaly $u$ determines the genus tower (the completion). The entire structure descends from a single identity: $D_{\mathcal{A}}^2 = 0$.

## 9. Concluding remarks

9.1. **The $E_n$ circle.** The holographic datum sits inside the $\mathsf{E}_n$ operadic circle:

$$\mathsf{E}_3^{\mathrm{top}}(\text{bulk}) \;\to\; \mathsf{E}_2(\text{boundary chiral}) \;\to\; \mathsf{E}_1(\text{bar/QG})$$
$$\to\; \mathsf{E}_2(\text{Drinfeld centre}) \;\to\; \mathsf{E}_3^{\mathrm{top}}(\text{derived centre}).$$

Each arrow is: restriction to codimension-2 defect, ordered bar complex, categorified averaging (Drinfeld centre), higher Deligne (derived centre). The circle partly closes: for simple $\mathfrak{g}$, the $\mathsf{E}_3$-topological deformation space of $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ is $\hbar\, H^3(\mathfrak{g})[[\hbar]]$, one-dimensional; the formal disk restriction recovers the Costello–Francis–Gwilliam Chern–Simons $\mathsf{E}_3$-algebra. The passage from $\mathrm{SC}^{\mathrm{ch,top}}$ to $\mathsf{E}_3$ requires a conformal vector at non-critical level (Sugawara), which makes $\mathbb{C}$-translations $Q$-exact and topologises the holomorphic direction. Without conformal vector: stuck at $\mathrm{SC}^{\mathrm{ch,top}}$.

9.2. **Scope.** The topologisation $\mathrm{SC}^{\mathrm{ch,top}} \to \mathsf{E}_3^{\mathrm{top}}$ is proved for affine Kac–Moody algebras $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$ (where the Sugawara construction is explicit). For general chiral algebras with conformal vector (Virasoro, $\mathcal{W}$-algebras), the topologisation is cohomological but the chain-level statement is open. The proof is cohomological: for class **M** algebras, where chain-level data is essential, $\mathsf{E}_3$ may exist only on cohomology.

The chiral Hochschild cohomology $\mathrm{ChirHoch}^*(\mathcal{A})$ is concentrated in degrees $\{0, 1, 2\}$ with polynomial Hilbert series (Theorem H). This is cohomological amplitude, not virtual dimension. It is not $\mathbb{C}[\Theta]$, and it is not the Gelfand–Fuchs cohomology (which is infinite-dimensional).

9.3. **From the datum to physics.** The derived chiral centre IS the bulk of a real 3d HT gauge theory. This is the content of the BBL identification: the bar construction on $\mathcal{A}$ produces a coalgebra; the chiral Hochschild cochain complex of $\mathcal{A}$ (computed from the bar resolution) produces the bulk observables; the $\mathrm{SC}^{\mathrm{ch,top}}$ structure on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ encodes the open/closed interaction. The datum $\mathcal{H}(\mathcal{A})$ is the smallest package from which the full physical correspondence reconstructs.

The programme beyond this paper: the global BBL triangle (bulk determines boundary determines bulk, proved boundary-linear; full reconstruction open), the chain-level topologisation for class **M**, and the closing of the $\mathsf{E}_n$ circle via the Drinfeld centre identification $Z(\mathbf{U}_{\mathcal{A}}) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$.

## References


[1] A. Beilinson and V. Drinfeld, *Chiral Algebras*, AMS Colloquium Publications **51**, 2004.

[2] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory*, Vols. 1–2, Cambridge University Press, 2017–2021.

[3] K. Costello, D. Francis, and D. Gwilliam, An $E_3$-algebra from Chern–Simons theory, arXiv:2602.12412, 2025.

[4] K. Costello and D. Gaiotto, Vertex algebras and 4d $\mathcal{N} = 2$ theories, *JHEP* **2019**(06):018, 2019.
[5] V. Drinfeld and D. Kazhdan, On the structure of quantum groups, *Funct. Anal. Appl.* **23**(1989), 115–132.
[6] P. Etingof and A. Varchenko, Geometry and classification of solutions of the classical dynamical Yang–Baxter equation, *Comm. Math. Phys.* **192**(1998), 77–120.
[7] B. Feigin and E. Frenkel, Affine Kac–Moody algebras at the critical level and Gelfand–Dikii algebras, *Int. J. Mod. Phys.* A**7**(Suppl. 1A)(1992), 197–215.
[8] D. Gaiotto and J. Zeng, Sphere algebras, arXiv:2026.xxxxx, 2026.
[9] V. Ginzburg and M. Kapranov, Koszul duality for operads, *Duke Math. J.* **76**(1994), 203–272.
[10] A. Khan and J. Zeng, The holomorphic-topological Poisson sigma model, arXiv:2026.xxxxx, 2026.
[11] V. Knizhnik and A. Zamolodchikov, Current algebra and Wess–Zumino model in two dimensions, *Nuclear Phys.* B**247**(1984), 83–103.
[12] M. Livernet, Koszulity of the category of Swiss-cheese operads, *J. Pure Appl. Algebra* **204**(2006), 475–495.
[13] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren **346**, Springer, 2012.
[14] R. Lorgat, *Modular Koszul Duality*, Vols. I–III, 2026.